Proofs

Loveless, Nix, Stokes, Abernathy's

May 28, 2026

Theorem. *The system of ordinary differential equations has a unique positive equilibrium point that is globally stable.*

Consider the system of ODEs

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + \frac{a_2 Ca}{Ca+\sigma} - r_1 A_\beta \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca. \end{cases}$$

Invariant. First we evaluate $\frac{dA_\beta}{dt}\Big|_{A_\beta=0}$ and $\frac{dCa}{dt}\Big|_{Ca=0}$. When $A_\beta = 0$, $\frac{dA_\beta}{dt} = a_1 + \frac{a_2 Ca}{Ca+\sigma}$. In our simulations we choose $a_1, a_2, \sigma > 0$, so $\frac{dA_\beta}{dt} > 0$ if $Ca \geq 0$. Similarly, when $Ca = 0$, $\frac{dCa}{dt} = b_1 + b_2 A_\beta$. Again, $b_1, b_2 > 0$ by choice, so $\frac{dCa}{dt} > 0$ when $A_\beta \geq 0$. Our model only pertains to real-world scenarios where A_β and Ca take non-negative values, thus $\frac{dA_\beta}{dt}\Big|_{A_\beta=0}, \frac{dCa}{dt}\Big|_{Ca=0} > 0$ hence the non-negative orthant is invariant.

Boundedness. Consider $\frac{dA_\beta}{dt}$. Notice that

$$a_1 + \frac{a_2 Ca}{Ca+\sigma} - r_1 A_\beta \leq a_1 + a_2 - r_1 A_\beta$$

since $\frac{Ca}{Ca+\sigma} \in [0, 1)$ for our chosen values of Ca. It follows that $a_1 + a_2 - r_1 A_\beta < 0$ when $A_\beta > \frac{a_1+a_2}{r_1}$. So, A_β is bounded. Then, choose some value M such that $M > A_\beta(t) \, \forall \, t \geq 0$. Now consider $\frac{dCa}{dt}$. Since A_β is bounded, we have

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \leq b_1 + b_2 M - r_2 Ca$$

So, $b_1 + b_2 M - r_2 Ca < 0$ when $Ca > \frac{b_1 + b_2 M}{r_2}$. Thus Ca is also bounded.

Equilibria. By solving $\frac{dA_\beta}{dt} = a_1 + \frac{a_2 Ca}{Ca+\sigma} - r_1 A_\beta$ for A_β we get

$$A_\beta = \frac{a_1 Ca + a_2 Ca + a_1 \sigma}{r_1 (Ca + \sigma)}.$$

We then substitute this value into $\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca$. The resulting expression is,

$$b_1 - r_2 Ca + \frac{b_2(a_1 Ca + a_2 Ca + a_1\sigma)}{r_1(Ca + \sigma)}.$$

Setting this expression equal to 0 and doing further simplification yields the quadratic,

$$-(r_1 r_2)Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma)Ca + a_1 b_2 \sigma + b_1 r_1 \sigma = 0.$$

Notice that the quadratic term is always negative and the constant term is always positive. The linear term can either be positive or negative depending on the choice of parameters. Regardless of the parity of the linear term, there is only one sign change in the expression, so by Descartes' Rule of Signs, there is one positive real root to this quadratic. Thus our system has one positive equilibrium point.

Non-existence of limit cycles. Let $\mathbf{F}(A_\beta, Ca) = \left\langle a_1 + \frac{a_2 Ca}{Ca+\sigma} - r_1 A_\beta, b_1 + b_2 A_\beta - r_2 Ca \right\rangle$. Then, $\nabla \cdot \mathbf{F} = -r_1 - r_2$. Choose the function $\alpha(A_\beta, Ca) = 1$. So, $\nabla \cdot (\alpha(\cdot)\mathbf{F}(\cdot)) < 0$, therefore by the Bendixson-Dulac Theorem, there is no limit cycle in the system of ODEs. So, the equilibrium point must be globally stable in the non-negative orthant.